OPG (parameter-golf): An Inference-Scaling Fixed-Point Mixture-of-Experts Language Model

Goal, theory, design, and current results

Project document for `train_gpt.py`

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Abstract

We develop a small language model whose central design objective is *inference-scaling*: spending more test-time compute — more steps of a recurrent solve — should yield better predictions, and should do so adaptively, spending more on harder inputs. Parameters are reused by tying one shared transformer-like block and iterating it; the expressiveness lost to that tying is recovered by a dense soft mixture of experts, which supplies a combinatorially large family of effective per-step weights — experts acting as a learned weight basis. The approach we pursue toward inference-scaling is a *reversible deep-equilibrium* (RevDEQ) solve: the model's representation is the fixed point of the shared block, a well-defined infinite-depth limit that a reversible backward pass trains in $O(1)$ activation memory regardless of solve depth.

This document is organized so that the theory stands on its own, independent of any particular training run. Section 1 states the goal and constraints; Section 2 positions the design against the recurrent-depth literature and inherits its evaluation axes; Section 3 develops a self-contained mathematical framework — the *contraction spectrum* — organized around a single control variable, the per-input Jacobian spectral radius $\rho(x)$ of the iteration map, and proves four results (a weight-tying expressiveness gap, a mixture-of-experts depth-recovery bound, a local-stability theorem, and an adaptive inference-scaling theorem) together with the evaluation metrics they justify; Section 4 describes the current architecture and training setup and ties each component to the theorem it serves; Section 5 reports the current pipeline's measured values and is explicit about the gap to the theoretical targets. The framework predicts that inference-scaling lives at criticality ($\rho(x) \to 1^-$); the current pipeline sits in the strongly contractive regime ($\rho \ll 1$, effective depth ≈ 1), so inference-scaling is *not yet realized* — this gap, and the metrics needed to close it, are the subject of the closing section.

Contents

1 Problem and Goal

1.1 Setting and measurement

We target a small, parameter-efficient language model and measure it by validation bits-per-byte (BPB) on a held-out language-modeling corpus (FineWeb, with a 1024-entry SentencePiece vocabulary). All development numbers in this document are controlled ∼1000-step runs that vary a single factor at a time, so comparisons are made at equal training-step count unless noted otherwise.

1.2 Goal: inference-scaling from a small parameter count

A large, deep, dense transformer is expressive because it has many distinct layers, each with its own weights; its parameter count grows linearly with depth. Our goal is to recover that expressiveness *without* paying for the depth in parameters, and to do so in a way that scales with test-time compute. Concretely the design rests on two complementary levers and one organizing objective.

Lever 1 — recurrent depth (parameter efficiency). We tie the weights of one shared block b_θ and obtain depth by *iterating* it. Depth is then a runtime quantity (how many times we apply b_θ), not a parameter-count quantity. This is the standard recurrent-depth / looped-transformer route to small models (Universal Transformer [4], Parcae [5], Huginn [6]).

Lever 2 — mixture-of-experts (expressiveness recovery). Tying a single block strictly reduces the function class relative to using independent per-layer weights (made precise in Theorem 3.3). We recover the lost expressiveness by making the shared block a *dense soft mixture of E experts*. At each application the block computes a token-dependent convex combination $\sum_e p_e(z)\, M_e(z)$ of the experts. The experts act as a learned *weight basis* and the routing weights $p(z)$ as *coordinates*: one shared block can realize a combinatorially large set of distinct effective weights, and — if its routing varies across iterations — the K-step recurrence can emulate the K distinct layers of a deep dense network (Theorem 3.5). This is the expressiveness-recovery role that MoE plays for layer-shared transformers in MoEUT [7] and the sparse-looped-MoE line.

Objective — inference-scaling. The property we ultimately require is *inference-scaling*: increasing the test-time compute budget (the recurrence length K) should not degrade and should, in the useful regime, *improve* accuracy, and the useful budget should be *larger for harder inputs*. This is the test-time-compute-scaling property of Huginn [6], Mixture-of-Recursions [8], and Think-at-Hard [9]. Inference-scaling — not merely "reach a stable state" — is the goal against which the design is judged.

Approach — the fixed point. The approach we pursue toward that goal is to define the model's representation as the *fixed point* of the shared block: for input embeddings x_0, $z^\star = T_\theta(z^\star, x_0)$. Three properties make the fixed point an attractive vehicle for inference-scaling. (i) It is a well-defined $K \to \infty$ limit, so "run the model longer" has a precise meaning. (ii) The number of iterations required to reach it is *input-dependent* — governed by how contractive the map is at that input — which is exactly the adaptive-compute behaviour we want (Section 3, Theorem 3.11). (iii) A reversible solver trains the limit in $O(1)$ activation memory, independent of depth (Section 4.4).

We are explicit, throughout, that a fixed point is the *means* and not the end. A strictly contractive map that converges *uniformly fast* for all inputs would plateau immediately and exhibit *no* inference-scaling. The theory in Section 3 therefore treats the contraction rate as a spectrum and identifies where on that spectrum inference-scaling lives, and where one would have to leave the fixed-point regime altogether.

1.3 Design requirements

Two requirements follow from the goal itself, not from any external budget:

- **Resource-constrained training and inference.** The model should train and run within the memory of a single commodity GPU (development uses one 48 GB L40S-class card); the reversible solver (Section 4.4) makes deep solves feasible there by keeping activation memory $O(1)$ in the solve depth.
- **Reversibility.** The solve must remain exactly invertible, so all routing is dense and soft — hard top-k dispatch, capacity dropping, and argmax selection are forbidden inside the solved block (Section 4).

2 Related Work and Baselines

The design sits inside the recurrent-depth / equilibrium-model family: models that trade extra compute for fewer unique parameters. We organize the comparison set by the mechanism each uses for depth and by whether it targets a fixed point or open-ended test-time computation, and we inherit several evaluation axes from this literature (Section 2.3). Replication state for each baseline is tracked in `experiments/docs/recurrent_depth_baselines.md` and the `baselines/` workspace.

2.1 Comparison set

- **DEQ / TorchDEQ** [1, 2]. The original deep-equilibrium models: solve for a fixed point of one block rather than stacking layers. Convergence is treated through implicit fixed-point solvers; our backbone is a reversible DEQ variant.
- **RevDEQ** [3]. Reversible DEQ with exact gradients and fewer function evaluations — the method closest to ours, which we extend with dense expert routing, Parcae damping, and explicit spectral diagnostics.
- **Universal Transformer** [4]. The seminal shared-block-through-depth model, optionally with adaptive per-position halting — an early adaptive-compute comparator.
- **Parcae** [5]. Stable looped language models via per-dimension damping and input injection, treating loop depth as a separate scaling axis; we adopt its damping in our solver.
- **Huginn / recurrent-pretraining** [6]. A large latent-recurrent LM with *variable test-time compute* — the canonical inference-scaling comparator.
- **MoEUT** [7]. Sparse mixture-of-experts capacity to recover the expressivity lost under Universal-Transformer-style layer sharing — the direct precedent for our Lever 2.
- **Mixture-of-Recursions** [8] and **MoDR**. Adaptive *token-level* recursion depth: route tokens to different numbers of recurrent passes — the adaptive-depth comparator most aligned with our goal.
- **Think-at-Hard** [9]. Selective latent iterations for *hard tokens* — explicitly "compute more where it is harder."
- **Iso-Depth Looped-LM Scaling** [10]. Measures the value of recurrence via a *recurrence-equivalence exponent* ϕ that separates true loop capacity from token-budget effects — a metric we inherit (Section 2.3).

2.2 Spectrum positioning

The framework of Section 3 places these models on a single axis: the per-input contraction rate $\rho(x)$ of the iteration map (formally Definition 3.1). Two camps emerge. The *equilibrium camp* (DEQ, TorchDEQ, RevDEQ, Parcae) drives the iteration to a fixed point; it gains a well-defined deep limit and stability, but a strictly converged solve plateaus and does not scale with extra test-time compute. The *test-time-scaling camp* (Huginn, Mixture-of-Recursions, Think-at-Hard) keeps the recurrence doing useful work — effectively operating away from a trivial equilibrium — and gains open-ended scaling at the cost of a convergence guarantee. This project aims to *bridge* the two: an expressive fixed point whose *approach* is input-adaptive, so that easy inputs converge quickly and hard inputs spend more iterations to reach a richer equilibrium. Table 1 summarizes the positioning.

Model	**Depth mechanism**	**Convergence guarantee**	**Test-time scaling**	**Expressivity recovery**
DEQ / TorchDEQ	implicit fixed point	solver residual	no (plateaus at FP)	none (single block)
RevDEQ	reversible fixed point	solver residual	no (plateaus at FP)	none (single block)
Universal Transformer	shared block $\times K$, ACT halting	none	adaptive halting	none
Parcae	damped looped block	stability via damping	loop-depth scaling law	none
Huginn	latent recurrence	none (by design)	**yes, variable**	large latent state
MoEUT	shared block + sparse MoE	none	no	**sparse MoE**
Mixture-of-Recursions / MoDR	per-token recursion depth	none	**yes, adaptive**	depth routing
Think-at-Hard	selective hard-token iterations	none	**yes, hard-token**	selective depth
This work	**reversible FP of dense-MoE block**	$\rho(J_F) < 1$ **(spectral)**	**target: adaptive via** $\rho(x) \to 1^-$	**dense soft MoE (weight basis)**

Table 1: Positioning against the recurrent-depth / equilibrium literature. Our design is the only entry combining a reversible expressive fixed point with a spectral convergence guarantee *and* an explicit inference-scaling target; the contraction rate $\rho(x)$ (Section 3) is the axis that unifies the "convergence guarantee" and "test-time scaling" columns.

2.3 Inherited evaluation axes

Several evaluation axes are imported from these comparators and instantiated as metrics in Section 3.7:

- **Loss-vs-compute scaling curve** (Huginn, Mixture-of-Recursions): BPB as a function of the test-time recurrence budget K. Our K-sweep is the population form of this curve; the per-input form is the new effective-depth metric of Section 3.7.
- **Recurrence-equivalence exponent** ϕ (Iso-Depth [10]): the exponent relating a looped model's effective capacity to an equivalent number of unique layers, separating genuine loop value from token-budget effects.
- **Fixed-point residual / convergence** (DEQ, RevDEQ): the relative change of the iterate at the deepest solve depth, our `iter_conv_rel`.
- **Expressivity-under-sharing** (MoEUT): whether the expert capacity is actually exercised, measured here through routing entropy, output orthogonality, and the across-depth routing variation.

3 Evaluation Metrics and Theoretical Guarantees

This section is mathematical and self-contained. It defines the objects, states the goal formally, proves the guarantees the design relies on, and derives the evaluation metrics from those guarantees. Nothing here depends on any particular training run; measured values appear only in Section 5.

3.1 Setup and notation

Let D be the model width and V the vocabulary size. An input is a token sequence x; write $x_0 \in \mathbb{R}^D$ for a single token's input embedding (state tensors carry batch and sequence axes, suppressed here). The shared block defines a parameterized map $T_\theta(\cdot, x_0) : \mathbb{R}^D \to \mathbb{R}^D$ for each fixed input x_0. The block has the input-injected form

$$T_\theta(z, x_0) = \bar{B} \odot \mathrm{RMSNorm}(x_0) + \Delta_\theta(z, x_0), \tag{1}$$

where $\bar{B}$ is a learned per-dimension input gain and Δ_θ is the expert-mixture update of Section 4. The model is solved by a damped two-state iteration (Section 4.4) whose joint update we write as $F(\cdot, x_0)$; its fixed point $z^\star(x_0)$ satisfies $z^\star = T_\theta(z^\star, x_0)$. The head $g : \mathbb{R}^D \to \mathbb{R}^V$ maps the solved representation to next-token logits, and the task loss at depth K is $L_K(x) = \ell(g(z_K(x)), y)$ with ℓ the cross-entropy (the basis of BPB). Write $L_\infty(x) = \ell(g(z^\star(x)), y)$ for the loss of the *defined* (fixed-point) model.

Definition 3.1 (Per-input contraction rate)**.** *For input x let $J_F(x) = \nabla_{(y,z)} F\big(z^\star(x), z^\star(x); x_0(x)\big)$ be the Jacobian of the joint iteration map at the fixed point, and define the* contraction rate

$$\rho(x) \;=\; \rho\big(J_F(x)\big) \;=\; \max\{|\lambda| : \lambda \textit{ eigenvalue of } J_F(x)\}.$$

The contraction rate is the single control variable of the framework: $\rho(x)$ governs both whether the solve converges (Theorem 3.7) and how many iterations it takes (Theorem 3.11).

Definition 3.2 (Inference-scaling)**.** *A recurrent model* exhibits inference-scaling *if (i) the expected loss $\mathbb{E}_x[L_K(x)]$ is non-increasing in the compute budget K over the operating range, and (ii) the budget $K^\star(x)$ required to reach a fixed loss tolerance is larger for harder inputs, where hardness is any input statistic monotone in $\rho(x)$ or in the initial distance $\|z_0(x) - z^\star(x)\|$. Property (i) is* no-degradation plus monotone improvement*; property (ii) is* adaptivity*.*

3.2 T1 — The weight-tying expressiveness gap

We first make precise why a tied recurrent block needs expressiveness recovery.

Theorem 3.3 (Weight-tying strictly shrinks the function class)**.** *Fix a block family $\{b_\theta : \theta \in \Theta\}$ with $\Theta \subseteq \mathbb{R}^p$ open and $b_\theta(\cdot)$ jointly smooth in (θ, z). For depth $L \geq 2$ define the independent-layer class $\mathcal{F}_L^{\mathrm{ind}} = \{b_{\theta_L} \circ \cdots \circ b_{\theta_1} : \theta_i \in \Theta\}$ and the tied class $\mathcal{F}_L^{\mathrm{tie}} = \{b_\theta^{\circ L} : \theta \in \Theta\}$. Then $\mathcal{F}_L^{\mathrm{tie}} \subseteq \mathcal{F}_L^{\mathrm{ind}}$, and the inclusion is strict for generic b: $\mathcal{F}_L^{\mathrm{tie}}$ is the image of a p-dimensional parameter manifold, while $\mathcal{F}_L^{\mathrm{ind}}$ is the image of an Lp-dimensional one, so $\mathcal{F}_L^{\mathrm{tie}}$ is a measure-zero subset of $\mathcal{F}_L^{\mathrm{ind}}$ whenever the independent parameterization has rank $> p$ at some point.*

Proof. Inclusion: set $\theta_i = \theta$ for all i. Strictness: the tied map $\theta \mapsto b_\theta^{\circ L}$ factors through the diagonal $\theta \mapsto (\theta, \ldots, \theta) \in \Theta^L$, so its image lies in a p-dimensional immersed submanifold of the function space spanned by the full $\theta_{1:L} \mapsto b_{\theta_L} \circ \cdots \circ b_{\theta_1}$ parameterization. If the latter has Jacobian rank $> p$ at any $\theta_{1:L}$ — which holds for generic smooth b with $L \geq 2$, since distinct layers can be perturbed independently — its image is a manifold of dimension $> p$, of which the p-dimensional diagonal image is a measure-zero subset. Hence there exist L-layer maps not realizable by any single iterated block. $\square$

Theorem 3.3 is the formal statement of "recurrent use of a shared block constrains performance." The remedy is to enlarge the per-step family so that *different applications can realize different effective layers* — the role of the mixture of experts.

3.3 T2 — Mixture-of-experts depth recovery

Let the block update be a dense soft mixture over E experts,

$$\Delta_\theta(z, x_0) \;=\; \sum_{e=1}^{E} p_e(z)\, M_e(z), \qquad p(z) \in \Delta^{E-1}, \tag{2}$$

with routing weights $p(z)$ on the probability simplex and experts M_e. The *effective per-step map* at routing p is $T_p(z) = \bar{B} \odot \mathrm{RMSNorm}(x_0) + \sum_e p_e M_e(z)$.

Proposition 3.4 (Per-step capacity: experts as a basis)**.** *Suppose each expert has a linear read-out, $M_e(z) = W_e^o\, \phi_e(z)$ with $W_e^o \in \mathbb{R}^{D\times r}$. Then for any state z the achievable update directions lie in* $\mathrm{span}\{W_1^o, \ldots, W_E^o\}$*, a subspace of dimension at most* $\min(\sum_e r,\ D)$*; if the read-outs are mutually orthogonal (e.g. a Cayley-parameterized stacked $[W_1^o \,\|\, \cdots \,\|\, W_E^o]$ with orthonormal columns), the achievable subspace has dimension exactly* $\min(Er, D)$*. In particular the function-space rank of one soft-MoE block is* $\min(Er, D)$*, recovering full dense width when $Er \geq D$.*

Proof. $\Delta_\theta(z, x_0) = \sum_e p_e(z) W_e^o \phi_e(z)$ is a linear combination of the columns of the W_e^o, hence lies in their joint span; its dimension is the rank of the stacked matrix, $\leq \min(Er, D)$, with equality under orthonormal columns. $\square$

Theorem 3.5 (Depth recovery from a single shared MoE block)**.** *Consider the K-fold composition of the shared block along a solve trajectory $z_{k+1} = T_{p(z_k)}(z_k)$, $k = 0, \ldots, K-1$.*

1. (Combinatorial layer count.) *Under an idealized hard routing that selects one expert per application, the K-step recurrence realizes one of up to E^K distinct ordered expert-sequences, i.e. up to E^K distinct depth-K composite maps from a single stored block. (With separate attention and MLP banks of E experts the count is $(E\cdot E)^K$.) Soft routing (2) is the differentiable convex relaxation of this selection, required for a reversible solve (Section 4).*
2. (Recovery condition.) *If the routing is constant along the trajectory, $p(z_k) \equiv p$, the composite collapses to $T_p^{\circ K}$, a single iterated map — back inside the tied class $\mathcal{F}_K^{\mathrm{tie}}$ of Theorem 3.3. The expressiveness of an independent-layer network is recovered only to the extent that the routing* varies across iterations*, i.e. only when the across-depth routing variation $\frac{1}{K-1}\sum_k \|p(z_{k+1}) - p(z_k)\|$ is bounded away from zero.*

Proof. (1) Each application chooses among E experts independently; the number of ordered length-K choices is E^K, each inducing a distinct composite for generic experts. The soft case replaces the choice by a point in Δ^{E-1}, a convex superset of the E vertices. (2) If $p(z_k) \equiv p$ then $z_{k+1} = T_p(z_k)$ for a fixed map T_p, so $z_K = T_p^{\circ K}(z_0) \in \mathcal{F}_K^{\mathrm{tie}}$; conversely distinct $T_{p(z_k)}$ across k produce a composition of distinct maps, which by Theorem 3.3 can lie outside the tied class. $\square$

Theorem 3.5 is the precise content of "MoE creates an exponential number of effective weights across depth that emulate the distinct layers of a dense deep transformer." Crucially it also identifies the *condition* for the recovery to occur — across-depth routing variation — which Section 3.7 turns into a directly measurable metric, and which Section 5 shows the current pipeline does not yet satisfy.

3.4 T3 — Local stability of the fixed point

The fixed point is the approach (Section 1.2); for it to be a well-defined, attracting limit we need the iteration to be locally contractive.

Definition 3.6 (Local exponential stability)**.** *Let $x^\star = M(x^\star)$ for an iteration $x_{k+1} = M(x_k)$. The fixed point is locally exponentially stable if there exist $\delta, C > 0$ and $\eta \in (0,1)$ with $\|x_0 - x^\star\| < \delta \Rightarrow \|x_k - x^\star\| \le C\eta^k \|x_0 - x^\star\|$ for all k.*

Theorem 3.7 (Spectral radius below one $\Rightarrow$ local exponential stability)**.** *Let M be C^1 near $x^\star = M(x^\star)$. If $\rho(\nabla M(x^\star)) < 1$, then $x^\star$ is locally exponentially stable, and for every $\eta \in (\rho, 1)$ there is C_η with $\|x_k - x^\star\| \le C_\eta\, \eta^k\, \|x_0 - x^\star\|$ near $x^\star$.*

Proof. Let $J = \nabla M(x^\star)$ with $\rho(J) < 1$. By the discrete Lyapunov theorem, for any $Q \succ 0$ there is a unique $P \succ 0$ with $J^\top P J - P = -Q$ (equivalently $P = \sum_{i\ge 0} (J^\top)^i Q J^i$, convergent since $\rho(J) < 1$). For the linearized error $e_{k+1} = J e_k$, $V(e) = e^\top P e$ obeys $V(Je) - V(e) = -e^\top Q e < 0$. For the nonlinear map, $M(x^\star + e) - x^\star = Je + r(e)$ with $\|r(e)\| = o(\|e\|)$; choosing a neighborhood with $\|r(e)\| \le \varepsilon \|e\|$,

$$V(Je + r) - V(e) \le -\lambda_{\min}(Q)\|e\|^2 + 2\|J^\top P\|\varepsilon\|e\|^2 + \|P\|\varepsilon^2\|e\|^2,$$

which is $\le -\frac{1}{2}\lambda_{\min}(Q)\|e\|^2$ for ε small. Hence V decreases geometrically, giving local exponential stability; the rate η can be taken arbitrarily close to $\rho(J)$ by the spectral radius formula $\rho(J) = \lim_k \|J^k\|^{1/k}$. □

Remark 3.8 (The relevant Jacobian is the two-state cycle F, not T_θ)**.** *The Parcae solver (Section 4.4) iterates a* joint two-state *map $F(y,z)$, not T_θ in isolation. With $D_A = \mathrm{diag}(\bar{A})$ and $D_\beta = \mathrm{diag}(1 - \bar{A})$ the per-dimension damping gates, and $J_T = \nabla T_\theta(z^\star)$,*

$$F(y,z) = \big(D_A y + D_\beta T_\theta(z),\ D_A z + D_\beta T_\theta(D_A y + D_\beta T_\theta(z))\big), \quad \nabla F = \begin{bmatrix} D_A & D_\beta J_T \\ D_\beta J_T D_A & D_A + D_\beta J_T D_\beta J_T \end{bmatrix}.$$

The relevant condition is therefore $\rho(J_F) < 1$, evaluated directly on F (Definition 3.1). Because D_A, D_β are per-dimension and do not commute with the dense J_T, damping alone cannot contract an expansive transition mode: the off-diagonal $D_\beta J_T$ blocks scale with $\|J_T\|$, so bringing $\rho(J_F) < 1$ requires joint *pressure on the damping* and *on the transition parameters. This is why $\rho(J_F)$ — not the damping coefficients, and not the transition map T_θ alone — is the principled gate.*

Remark 3.9 (Spectral radius, not operator norm)**.** *$\rho(J_F) < 1$ is* necessary and sufficient *for local exponential stability (Theorem 3.7 and its converse via $\rho = \lim \|J^k\|^{1/k}$). The operator-norm condition $\sigma_{\max}(J_F) < 1$ is* sufficient but not necessary*, and for non-normal J_F the gap is large: a Jacobian can satisfy $\rho \ll 1$ while $\sigma_{\max} \gg 1$ (transient amplification that decays geometrically). We therefore gate on $\rho(J_F)$ and report $\sigma_{\max}$ only as a diagnostic; a soft penalty on $\sigma_{\max}$ targets the wrong condition and is not used (Section 3.7).*

3.5 T4 — Adaptive inference-scaling on the contraction spectrum

We now connect the convergence metric $\rho(x)$ to the goal of Definition 3.2.

Assumption 3.10 (Local Lipschitz head)**.** *The composed head-and-loss $z \mapsto \ell(g(z), y)$ is L_{head}-Lipschitz on a neighborhood of $z^\star(x)$ containing the solve trajectory. (This holds for the architecture of Section 4:* RMSNorm *is smooth away from* 0*, the read-out is linear, and log-softmax cross-entropy is* 1*-Lipschitz in the logits in total variation.)*

Theorem 3.11 (Contraction rate controls the loss gap and the adaptive budget)**.** *Fix x with $\rho(x) < 1$ (so by Theorem 3.7 the solve converges locally) and let Assumption 3.10 hold. Then for every $\eta \in (\rho(x), 1)$ there is $C(x) > 0$ with*

$$\|z_K(x) - z^\star(x)\| \le C(x)\,\eta^K\,\|z_0(x) - z^\star(x)\|, \qquad |L_K(x) - L_\infty(x)| \le L_{\text{head}}\,C(x)\,\eta^K\,\|z_0(x) - z^\star(x)\|. \tag{3}$$

Consequently the compute required to bring the loss gap below a tolerance $\delta > 0$ is

$$K^\star(x) \;=\; \left\lceil \frac{\log\!\big(L_{\text{head}} C(x)\,\|z_0(x) - z^\star(x)\|/\delta\big)}{\log(1/\eta)} \right\rceil, \tag{4}$$

which (i) is finite and the loss is monotonically non-increasing for $K < K^\star(x)$, and (ii) increases as the contraction weakens *($\rho(x) \to 1^-$ forces $\eta \to 1^-$, so* $\log(1/\eta) \to 0^+$ *and $K^\star(x) \to \infty$)* and as the initial distance $\|z_0(x) - z^\star(x)\|$ grows*. Hence a family of fixed-point solves with input-dependent $\rho(x) < 1$ satisfies both clauses of Definition 3.2: monotone improvement to the equilibrium, with per-input budget increasing in hardness.*

Proof. Equation (3): the first bound is Theorem 3.7 applied to F (Remark 3.8), projected to the z-coordinate; the second composes it with Assumption 3.10. Equation (4): solve $L_{\text{head}}C(x)\eta^K\|z_0 - z^\star\| \le \delta$ for K. Monotonicity on $K < K^\star(x)$ follows because the right side of (3) is strictly decreasing in K; the limits in (ii) are immediate from $\log(1/\eta) \to 0^+$ as $\eta \to 1^-$ and the linear dependence on $\|z_0 - z^\star\|$. ◻

Corollary 3.12 (The contraction spectrum)**.** *Reading $K^\star(x)$ as the effective test-time depth, Theorem 3.11 partitions the design space by $\rho(x)$:*

- $\rho(x) \ll 1$ ***(strongly contractive):*** *$K^\star(x) \approx 1$ for all inputs — the solve settles immediately, there is* no *inference-scaling (effective depth ≈ 1).*
- $\rho(x) \to 1^-$ ***(critical):*** *$K^\star(x)$ is large and* heterogeneous*, growing with input hardness — this is the regime in which inference-scaling holds and, by Theorem 3.5(2), the long transient composes many distinct MoE layers.*
- $\rho(x) \ge 1$ ***(non-equilibrium):*** *no fixed point is reached; open-ended test-time computation is possible (the Huginn/MoR regime) but the convergence certificate of Theorem 3.7 is forfeited. This is the* relax-FP fork*: it is reachable without losing the $O(1)$-memory reversible training (which needs only floored invertibility, Section 4.4), but it trades the guarantee for the scaling.*

The design goal is to operate at criticality from below ($\rho(x) \to 1^-$) with $\rho(x)$ heterogeneous across inputs.

3.6 The explicit guarantee: optimizing the metric drives the goal

We close the loop by exhibiting a training objective whose minimization provably advances the premises of Theorem 3.11. Let $S_0 < S_1 < \cdots < S_m$ be solve depths along a single trajectory and define the recursive multi-K consistency loss

$$\mathcal{L}_{\text{consist}} = \frac{1}{m} \sum_{i=0}^{m-1} \left\| z_{S_i} - \text{sg}(z_{S_{i+1}}) \right\|^2. \tag{5}$$

Proposition 3.13 (Consistency bounds representation error and forces contraction)**.** *The consistency loss upper-bounds the trajectory's terminal displacement, $\|z_{S_0} - z_{S_m}\| \le \sum_i \|z_{S_i} - z_{S_{i+1}}\| \le \sqrt{m}\left(\sum_i \|z_{S_i} - z_{S_{i+1}}\|^2\right)^{1/2}$, so driving $\mathcal{L}_{\text{consist}} \to 0$ makes the trajectory Cauchy and hence (with a fixed point present) drives the representation error $\|z_{S_m} - z^\star\| \to 0$, the premise of Theorem 3.11. Moreover $\mathcal{L}_{\text{consist}} = 0$ on a non-stationary trajectory is achievable only if the iterates have stopped moving, i.e. only if the map is contractive at the operating point ($\rho(J_F) < 1$); the recursive (consecutive-pair) form, unlike an all-to-deepest target, only requires* local *contraction over each depth range $[S_i, S_{i+1}]$ and so does not collapse the map to a trivial $\rho \to 0$ degenerate solution.*

Proof. The displacement bound is the triangle inequality followed by Cauchy–Schwarz; a Cauchy solve with a fixed point converges to it. For the contraction statement, if consecutive iterates coincide then $z_{S_{i+1}} = F(z_{S_i}) \Rightarrow z_{S_i}$ is fixed, which near a hyperbolic fixed point requires $\rho(J_F) < 1$ (Theorem 3.7); the locality of the consecutive-pair constraint is immediate since each term involves only adjacent depths. □

What is proven, and what is open. Proposition 3.13 with Theorems 3.7–3.11 establishes the following honest chain: *optimizing* the consistency objective and the $\rho(J_F)$ gate *provably* yields a well-defined, locally stable fixed-point model whose finite-K solve is a faithful surrogate for the infinite-depth limit (loss gap (3)), with the adaptive-budget structure (4) required for inference-scaling, and a per-step expressiveness floor $\min(Er, D)$ (Proposition 3.4). Three things are *not* proven and are flagged as open: (a) the *absolute* quality of the fixed-point predictor (no universal-approximation result is claimed — expressiveness is a capacity floor, not a guarantee of fit); (b) *global* convergence from arbitrary initialization (the theory is local at the reached equilibrium); and (c) that training can be driven to the *critical, heterogeneous* $\rho(x) \to 1^-$ regime — the consistency loss bounds representation error and enforces $\rho < 1$, but it does not by itself push ρ toward 1 from below, which is the central tension between expressiveness and contraction (Corollary 3.12; Section 5.5).

3.7 Metrics derived from the theory

Each metric is justified by the theorem it gates and is tagged by instrumentation status: **[emitted]** = computed every evaluation by the current pipeline; **[target]** = required by the goal but not yet instrumented (with the instrumentation it needs).

Metric	What it measures	Gated by	Status / instrumentation
Objective			
BPB	bits/byte on full validation, post-quant	— (the score)	**[emitted]**
Convergence (T3, Thm. 3.7)			
$\rho(J_F)$	spectral radius of the iteration Jacobian, per K	Thm. 3.7 (nec. & suff.)	**[emitted]** (power iteration on J_F)
$\sigma_{\max}(J_F)$	operator norm of the Jacobian	reported only (Rmk. 3.9)	**[emitted]**, not gated
`iter_conv_rel`	relative iterate change at deepest K	empirical conv. (< 0.05)	**[emitted]**
`deq_recon_err`	forward distance travelled (TBPTT)	not a recon error	**[emitted]**, interpret per def.
Inference-scaling (T4, Thm. 3.11; Def. 3.2)			
loss-vs-K curve	BPB vs. test-time budget K (population)	Def. 3.2(i)	**[emitted]** (the K-sweep)
eff. depth $K^\star(x)$ dist.	per-input iterations-to-tolerance	Def. 3.2(ii)	**[target]**: log per-token convergence step within the solve
$\rho(x)$ heterogeneity	spread of contraction rate over inputs	Cor. 3.12	**[target]**: bucketed power-iteration of $J_F(x)$
ϕ (recurrence-equiv.)	loop value vs. unique-layer count [10]	Def. 3.2	**[target]**: fit depth-vs-loss across K
Expressiveness (T1/T2, Thm. 3.3/3.5)			
across-depth routing var.	$\frac{1}{K-1}\sum_k \Vert p(z_{k+1}) - p(z_k)\Vert$	Thm. 3.5(2)	**[emitted]** as iteration-variation
output orthogonality	cosine between expert outputs	Prop. 3.4 (basis rank)	**[emitted]**
per-token entropy	routing concentration per token	specialization	**[emitted]**
utilization entropy	usage spread across the batch	whether all experts used	**[emitted]**
Health and budget			
load-balance CV; min share	balance; dead-expert detection	routing health	**[emitted]**
artifact bytes; quant gap	compressed size; cost of int6	deployability	**[emitted]**
peak VRAM; param count	single-GPU fit; model size	constraint	**[emitted]**
NTP descent rate	per-step & per-wallclock loss slope	training health	**[emitted]**

Table 2: Theory-driven metric set. The convergence and expressiveness metrics are emitted today; the inference-scaling metrics ($K^\star(x)$ distribution, $\rho(x)$ heterogeneity, ϕ) are the targets that directly measure the goal and are not yet instrumented. Promotion gate: $\rho(J_F) < 1$ (theoretical) or `iter_conv_rel`< 0.05 at deepest K (empirical).

4 Current Design and Training Setup

`train_gpt.py::Hyperparameters` is the source of truth; this section mirrors it and states, for each component, which theorem of Section 3 it serves.

4.1 Backbone and the fixed-point map

For token embeddings the input state is $x_0 = \mathrm{RMSNorm}(E[x])$ (BigramHash disabled by default). For a tensor u, $\mathrm{RMSUnit}(u) = u/\sqrt{D^{-1}\sum_i u_i^2 + \epsilon}$ with $\epsilon = 10^{-6}$, and a learned scale gives $\mathrm{RMSNorm}(u; g) = \mathrm{RMSUnit}(u) \odot g$. The block uses the parameter-free preconditioner $h = \mathrm{RMSUnit}(z + x_0)$ and forms the expert-mixture update $\Delta_\theta(z, x_0) = \Delta_\theta^A + \Delta_\theta^M$ (attention and MLP banks, Section 4.3). The fixed-point map is the input-injected form (1): $T_\theta(z, x_0; \bar{B}) = \bar{B} \odot \mathrm{RMSNorm}_{x_0}(x_0) + \Delta_\theta(z, x_0)$, with learned input-injection scale $\mathrm{RMSNorm}_{x_0}(x_0) = \mathrm{RMSUnit}(x_0) \odot g_{x0}$. There is no unconditional residual $x_0 + \Delta_\theta$: the input conditions the map both structurally (through h) and explicitly (through $\bar{B}$). *Serves:* defines $z^\star$ as the infinite-depth limit (Section 1.2), the object of Theorem 3.7.

4.2 Dense soft-routed mixture of experts (Lever 2)

All experts process all tokens; the router forms continuous mixture weights. The default scorer is evidential Dirichlet-UCB: per token, $\alpha = \mathrm{softplus}(\ell(h)) + 1$ defines a Dirichlet posterior with mean $\mu = \alpha/\sum \alpha$ and per-expert standard deviation σ_e, and routing weights are $p = \mathrm{acq}/\sum \mathrm{acq}$ with $\mathrm{acq} = \mu + \beta\sigma$, $\beta = 0.5$ annealing to zero. The routed sigmoid gate is disabled by default, so $\sum_e p_e = 1$ exactly. One pooled router over $2R$ outputs routes the R attention and R MLP experts ($R = E$ with no shared experts). *Serves:* Theorem 3.5 (the weight basis) and Proposition 3.4 (rank $\min(Er, D)$). Routing must be soft and dense: hard top-k, capacity, and argmax are discrete inside T_θ and break the reversible solve, and are rejected by startup validation.

4.3 Independent experts: MLA attention and SwiGLU MLP

Each attention expert owns a full multi-head latent-attention (MLA) pipeline: low-rank query projection with per-head gate logits, low-rank KV compression, independent K/V decompression, independent low-rank RoPE key projection, per-expert Q/K RMS scales, and a low-rank output projection (attention rank 64). Expert index is packed into the head dimension so one SDPA call with grouped-query attention evaluates all experts. Each MLP expert is a full-D low-rank SwiGLU bank, $r_e(h) = \mathrm{SiLU}(W_e^g(h \odot g_e^g)) \odot W_e^f(h \odot g_e^f)$, $M_e(h) = W_e^o(\mathrm{RMSUnit}(r_e(h)) \odot g_e^o)$ (MLP rank 96, `mlp_mult` $= 3$). Every expert-path parameter is per-expert; no learned scale is shared across experts. *Serves:* the orthogonalizable read-outs of Proposition 3.4 (the iter175 Cayley-orthogonal read-out, queued, would make the basis exactly orthonormal).

4.4 Parcae damping and the reversible solve (the approach)

Raw Parcae parameters produce positive magnitudes δ, a, b (softplus+10^{-3}), and with reversibility floor $\epsilon_{\mathrm{rev}} = 0.1$,

$$\bar{A} = \epsilon_{\mathrm{rev}} + (1 - \epsilon_{\mathrm{rev}}) \exp(-\delta \odot a), \qquad \beta = 1 - \bar{A}, \qquad \bar{B} = \delta \odot b,$$

so $\bar{A} \in (\epsilon_{\mathrm{rev}}, 1)$ (init ≈ 0.7) and $\bar{B}$ is the input gain (init ≈ 0.3). The forward solve runs the two-state damped iteration from $y_0 = z_0 = x_0$,

$$y_{k+1} = \bar{A} \odot y_k + \beta \odot T_\theta(z_k, x_0; \bar{B}), \qquad z_{k+1} = \bar{A} \odot z_k + \beta \odot T_\theta(y_{k+1}, x_0; \bar{B}),$$

which at a common equilibrium reduces to $z^\star = T_\theta(z^\star, x_0; \bar{B})$. The update is exactly invertible: $z_k = (z_{k+1} - \beta \odot T_\theta(y_{k+1}))/\bar{A}$ and $y_k = (y_{k+1} - \beta \odot T_\theta(z_k))/\bar{A}$, evaluated in fp64; the backward pass *reconstructs* states from the terminal state, so activation memory is $O(1)$ in K. The floor on $\bar{A}$ is a correctness constant (the reverse denominator), not a tuning knob; $\bar{B}$ is never a divisor and is not floored. *Serves:* Theorem 3.7/Remark 3.8 (the iterated map is F, whose damping sets the ρ axis) and the $O(1)$-memory claim that makes training the infinite-depth limit feasible on one GPU — a property that, as Corollary 3.12 notes, survives even the relax-FP fork since it needs only floored invertibility.

Training samples the forward depth K from `deq_k_jitter_set` $= \{16, 24, 32, 64, 96, 128\}$, with most mass on the shallow depths (K=16 and K=24 carry weights 0.50 and 0.40) and a light deep tail (K=96, 128 share $\approx 2\%$); evaluation defaults to K=16; the gradient is truncated to the last K_{bptt}=3 iterations.

4.5 Consistency anchors (the explicit-guarantee objective)

With `deq_prefix_anchors=True` (default), the task cross-entropy is averaged over every prefix anchor depth traversed within the single sampled-K solve, and the recursive multi-K consistency loss (5) anchors each gradient-carrying intermediate to its next-deeper neighbor over the depth set $S = (8, 16, 24, 32, 64, 128)$ with coefficient $\lambda_a =$ `multi_k_consistency_anchor_coef` $= 0.1$. The targets $z_{S_{i+1}}$ are produced by the same forward pass, so there is no extra forward compute. *Serves:* Proposition 3.13 — this is the objective whose minimization provably bounds the representation error and enforces $\rho(J_F) < 1$.

4.6 Head, losses, optimizer, quantization

Head. The output uses a low-rank mixture-of-softmaxes (MoS) over the solved representation $h_{\text{out}} = \text{RMSNorm}(z^\star)$ (two shared + one specialized component, pure softmax routing, FSQ disabled, NTP-only; CTP banks are not allocated by default). Its smoothness is the basis of Assumption 3.10.

Training loss. $\mathcal{L} = \mathcal{L}_{\text{NTP}} + \mathcal{L}_{\text{router}} + \mathcal{L}_{\text{consist}}$ (the Jacobian and denoising regularizers $\lambda_J = \lambda_D = 0$ by default). The router term is a flat stack: per-token entropy ($\lambda_{\text{ent}} = 0.1$), EMA-anchored balance via reverse KL ($\lambda_{\text{bal}} = 0.30$), EMA-anchored specialization ($\lambda_{\text{spec}} = 0.20$), a straight-through alive hinge ($\lambda_{\text{alive}} = 0.02$), and worst-pair cosine expert-output diversity ($\lambda_{\text{div}} = 0.30$), all ramped over the first 7% of training. Load CV is diagnostic-only (removed as a loss).

Optimizer. Matrix parameters use Muon with Polar-Express Newton–Schulz (7 backend steps, momentum 0.99, matrix lr 0.022); scalars and embeddings use Adam (β_1=0.85, β_2=0.90, tied-embed lr 0.03); Parcae raw parameters use a dedicated lr 0.002 with no weight decay. Global weight decay is 0.015 and the gradient-norm clip is 1.0.

Quantization and validation. Matrix tensors are int6 per-row quantized (token embeddings kept fp16) and zstd-22 compressed via `encode_scored_artifact` into a compact deployable artifact; the model is then dequantized and re-validated, and the post-quantization BPB is the reported score. The diagnostic eval profile sweeps $K \in \{4, 8, 16, 17, 24, 32, 37, 64, 113, 128, 192\}$ (primes 17, 37, 113 are acyclicity checks; 192 is an extrapolation point) and emits $\rho(J_F)$ on every row.

Default configuration table. (`num_layers`$= 12$ is the maximum solver depth K.)

field	default	field	default
`vocab_size`	1024	`train_seq_len`	2048
`train_batch_tokens`	524,288	`iterations`	1000
`num_layers` (max depth K)	12	`model_dim`	768
`num_heads`	8	`num_kv_heads`	4
`num_experts` (E)	16	`num_shared_experts`	0
`attn_expert_rank`	64	`mlp_expert_rank`	96
`mlp_mult`	3.0	`tie_embeddings`	true
`use_parcae`	true	`parcae_reversibility_floor`	0.1
`parcae_init_a_bar`	0.7	`parcae_init_b_bar`	0.3
`deq_k_jitter_set`	$(16, 24, 32, 64, 96, 128)$	`deq_k_eval`	16
`deq_bptt_k`	3	`router_scoring`	dirichlet_ucb
`deq_prefix_anchors`	true	`deq_prefix_anchor_set`	$(8, 16, 24, 32, 64, 128)$
`multi_k_consistency_anchor_coef`	0.1	`use_router_sigmoid_gate`	false
`router_ema_balance_coef`	0.30	`router_ema_specialization_coef`	0.20
`router_pertoken_entropy_coef`	0.1	`expert_output_diversity_coef`	0.30
`router_ema_alive_coef`	0.02	`regularizer_warmup_frac`	0.07
`use_ctp`	false	`num_refinements`	0
`lyapunov_coef`	0.0	`denoising_coef`	0.0
`matrix_lr`	0.022	`parcae_lr`	0.002
`weight_decay`	0.015	`grad_clip_norm`	1.0

Where the current design sits on the spectrum. By Corollary 3.12, the design choices above — Parcae damping initialized at $\bar{A} \approx 0.7$, a consistency loss that enforces $\rho < 1$ — place the model in the strongly contractive regime $\rho \ll 1$. The theory therefore *predicts* effective depth ≈ 1 and no inference-scaling for the current configuration, and identifies the lever that would move it toward criticality: increasing across-depth routing variation (Theorem 3.5(2)) so that the transient does genuine per-depth work while $\rho(x)$ rises toward 1 from below. Section 5 confirms this prediction empirically.

5 Experimental Results

The numbers below are a snapshot of the current pipeline (the iter172 development champion, a controlled ~1000-step run); the pipeline is still improving. They are reported against the metric set of Table 2, and the gap to the theoretical targets of Section 3 is made explicit in Section 5.5.

5.1 Headline metrics

Metric	Meaning	Value	Target / cap
BPB (primary)	full-validation, post-int6-quant	**1.462898**	lower is better
$\rho(J_F)$ @ K=128	spectral radius of J_F	0.77	<1 (contractive)
$\sigma_{\max}(J_F)$	operator norm (reported only)	≈ 14.5	not gated
conv. residual @ deepest K	relative iterate change	0.011	<0.05
K-extrap. drift	BPB from K=16 to K=128	+0.6 mBPB	flat
artifact size	int6 weights + code	7.40 MB (7,432,396 B; 7.96 MB w/ code)	—
quantization gap	fast proxy → full post-quant	+26 mBPB(1.4369→ 1.4629)	smaller
peak VRAM	deep solve, single GPU	36.9 GB	<48 GB
parameters	total	≈ 13.9 M	—

Table 3: Headline values for the current model. The convergence gate is met ($\rho(J_F) = 0.77 < 1$, residual $0.011 < 0.05$); the operator norm is large but ungated (Remark 3.9); the compressed int6 artifact is 7.40 MB and the deep solve fits one 48 GB GPU.

5.2 Behaviour across solver depth (the loss-vs-compute curve)

Table 4 sweeps fixed solver depths, the population form of the inherited loss-vs-compute curve. BPB is flat once $K \geq 16$ and the residual falls monotonically to ≈ 0.011, while the shallow solves ($K = 4, 8$) are still settling — the model has learned a genuine equilibrium, not a memorized step count. $\rho(J_F)$ is a power-iteration estimate, noisy at individual K but settling to ≈ 0.77 deep; $\sigma_{\max} \gg 1$ throughout, which is exactly why it is not the gate (Remark 3.9). The expert-health columns are essentially K-invariant — routing does not vary with eval depth, the direct signature of the effective-depth-≈ 1 regime (Section 5.5).

K	BPB	resid.	$\rho(J_F)$	$\sigma_{\max}$	$\mathrm{CV}_{a/m}$	$\mathrm{ortho}_{a/m}$	H_{tok}
4	1.6235	0.304	1.08	13.3	0.34/0.10	0.05/0.23	3.24
8	1.4745	0.076	0.79	14.0	0.32/0.10	0.05/0.23	3.25
16	1.4716	0.013	1.25	15.0	0.32/0.10	0.05/0.23	3.25
24	1.4720	0.011	0.87	14.6	0.32/0.10	0.05/0.23	3.25
32	1.4721	0.011	1.02	14.7	0.32/0.10	0.05/0.23	3.25
64	1.4722	0.011	0.77	14.3	0.32/0.10	0.05/0.23	3.25
128	1.4722	0.011	0.77	14.5	0.32/0.10	0.05/0.23	3.25

Table 4: Fixed-K post-quant diagnostic sweep (the headline full-validation BPB is 1.4629). "a/m" = attention / MLP routers; H_{tok} is per-token routing entropy. BPB is flat for $K \geq 16$; the expert metrics are K-invariant.

5.3 The convergence arc

The development trajectory shows BPB and convergence improving *together* (Table 5): the consistency signal first pulled $\rho(J_F)$ from a divergent ≈ 1.30 to 0.92 at near-flat BPB, then its recursive multi-K form pushed $\rho(J_F)$ to 0.77 and delivered the largest BPB gain. This is consistent with

Proposition 3.13 (consistency enforces $\rho < 1$) and with the loss-gap bound (3) (a more faithful FP surrogate lowers the trained loss); it is reported as an observed development arc, not as evidence for any theorem.

Stage	**bpb**	$\rho(J_F)$	**Change**
baseline	1.4718	$\approx 1.30^{\dagger}$	no convergence signal; solve not yet contractive
+ consistency signal	1.4716	0.92	convergence, not BPB, was the win
+ recursive multi-K	**1.4629**	**0.77**	-8.7 mBPB; current model

Table 5: Controlled $\sim$1000-step development runs. †Without the convergence signal the solve is not contractive; the signal trains $\rho(J_F)$ below 1.

5.4 Expert and routing health

All 16 experts are active and well utilized: load-balance CV 0.32/0.10 (attn/MLP), minimum per-expert share 3.9%/5.6% (no dead experts), per-token entropy 3.25, utilization entropy 3.44 (of a maximum 3.47, i.e. 99% of experts used), output orthogonality 0.05/0.23 (experts are distinct). By Proposition 3.4 the capacity floor is being exercised across experts; by Theorem 3.5(2) the relevant unmet quantity is across-depth variation (next subsection).

5.5 Gap to the theoretical targets

The framework's central prediction for the current configuration ($\rho \ll 1$, Corollary 3.12) is confirmed: the across-depth routing variation (iteration-variation) is 0.001–0.005, i.e. routing is essentially constant across solver iterations. By Theorem 3.5(2) the K-step composite therefore collapses toward a single iterated map (**effective depth** ≈ 1), and by Corollary 3.12 the strongly contractive solve yields a homogeneous $K^{\star}(x) \approx 1$ — so **inference-scaling is not yet realized**: extra test-time compute past $K = 16$ does not improve BPB (Table 4), and there is no evidence of harder inputs using more effective depth. Consistent with this, the three inference-scaling metrics that would *measure* the goal — the effective-depth distribution $K^{\star}(x)$, the per-input $\rho(x)$ heterogeneity, and the recurrence-equivalence exponent ϕ — are **not yet instrumented** (Table 2, status [**target**]). Today's expressiveness comes from the richness of the converged MoE map (Proposition 3.4), not from depth-specialization.

5.6 Open problems and next steps

1. **Instrument the inference-scaling metrics.** Add per-token convergence-step logging ($K^{\star}(x)$ distribution), bucketed $\rho(x)$ estimation, and a ϕ fit (Table 2); without them the goal is not directly measurable.
2. **Drive across-depth routing variation toward criticality.** Push the model from $\rho \ll 1$ toward $\rho(x) \to 1^{-}$ with $\rho(x)$ heterogeneous, so the transient composes many distinct MoE layers (Theorem 3.5(2)) and hard inputs use more depth (Theorem 3.11). This is the central expressiveness-vs-contraction tension flagged as open in Section 3.6.
3. **Strengthen the expert basis.** The queued E=64 Cayley-orthogonal read-out (Proposition 3.4) raises the per-step rank ceiling from 15-D to 63-D.
4. **Differentiated per-depth state (DeepSeek mHC).** Manifold-constrained hyper-connections — several parallel streams mixed by a magnitude-preserving doubly-stochastic matrix — are a

candidate source of the differentiated state that step 2 needs, and their magnitude preservation aligns with reversibility and $\rho(J_F) < 1$.

5. **Matched-baseline comparison.** Benchmark against the Section 2 set on the inherited axes (loss-vs-compute, ϕ).
6. **The relax-FP fork.** If criticality from below proves insufficient for the required scaling, deliberately enter the $\rho \geq 1$ non-equilibrium regime (Corollary 3.12) — retaining $O(1)$-memory reversible training but replacing the spectral certificate with an operational scaling guarantee.

A Implementation Checklist

Keep this document synchronized with `train_gpt.py` if any of the following change: the block equation (1); the Parcae coefficients or reversibility floor; the router scoring or pooled-router layout; the consistency objective (5) or its anchor set; the flattened router/diversity coefficients or warmup; the training timer semantics; the default CTP, refinement, Lyapunov, denoising, or TBPTT settings; the post-int6 validation gates or K-sweep values; the parameter count, artifact serialization, or eval profiles.

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